

# Query-GUI Language Matching in Multilingual GUI Understanding

다국어 GUI 이해에서 질문-인터페이스 언어 일치의 효과

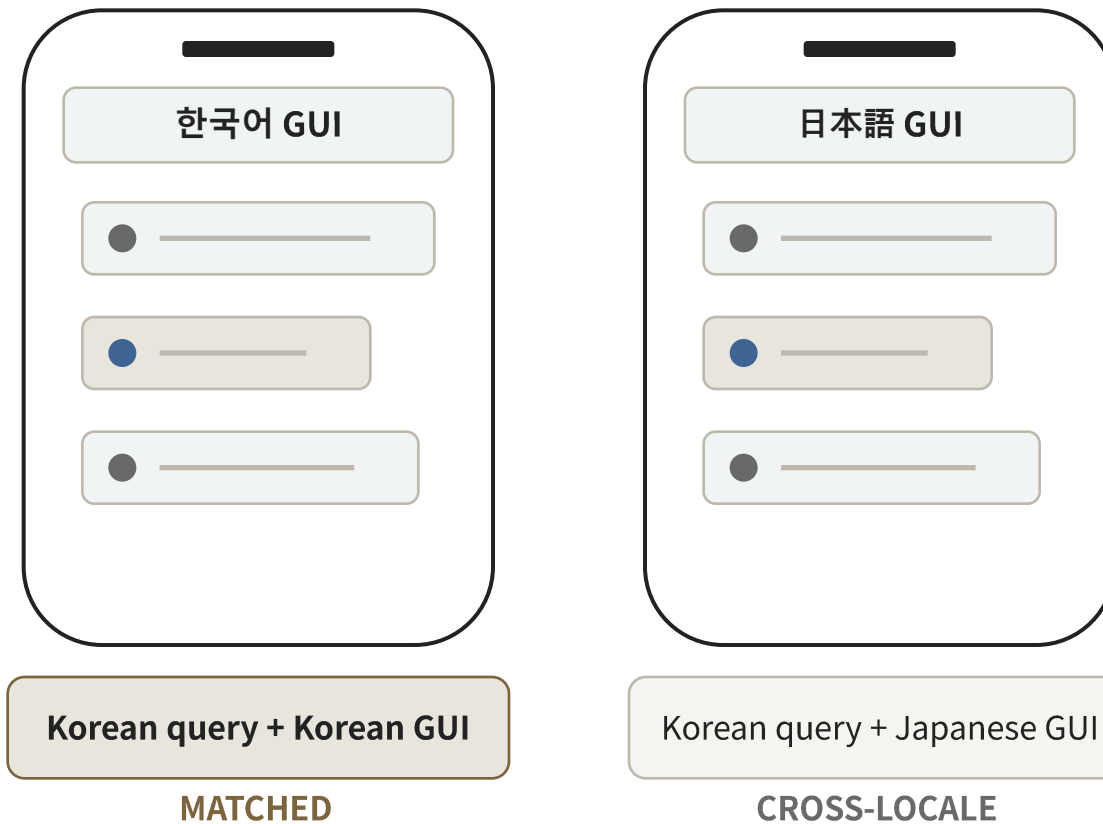
Does matching the query and GUI language improve multilingual GUI understanding?

이승재, 배기현 / MM 2팀

# Multilingual GUI 이해에서의 Query-GUI 언어 일치

Research motivation · separating concordance from language difficulty

## Same task, different language relation



## Why a raw accuracy gap is ambiguous

- Query-language difficulty
- GUI-language difficulty
- Query-GUI concordance
- Item-specific difficulty

- RQ1** 언어와 문항 난이도를 통제해도 concordance advantage가 남는가?
- RQ2** 이 차이는 GUI capability에 따라 달라지는가?

# Fully Crossed 6×6 Query–GUI Language Evaluation

The same semantic item is repeated across every language pair

|       |    | GUI language |    |    |    |    |    |
|-------|----|--------------|----|----|----|----|----|
|       |    | EN           | ZH | FR | RU | JA | TH |
| Query | EN | M            |    |    |    |    |    |
|       | ZH |              | M  |    |    |    |    |
|       | FR |              |    | M  |    |    |    |
|       | RU |              |    |    | M  |    |    |
|       | JA |              |    |    |    | M  |    |
|       | TH |              |    |    |    |    | M  |

Diagonal = matched

Off-diagonal = cross-locale

2

Models

2,195

Semantic Items

6

Query Languages

6

GUI Languages

79,020

Evaluations / model

$$2,195 \times 6 \times 6 = 79,020$$

Perception / Action · WF WI AU AP

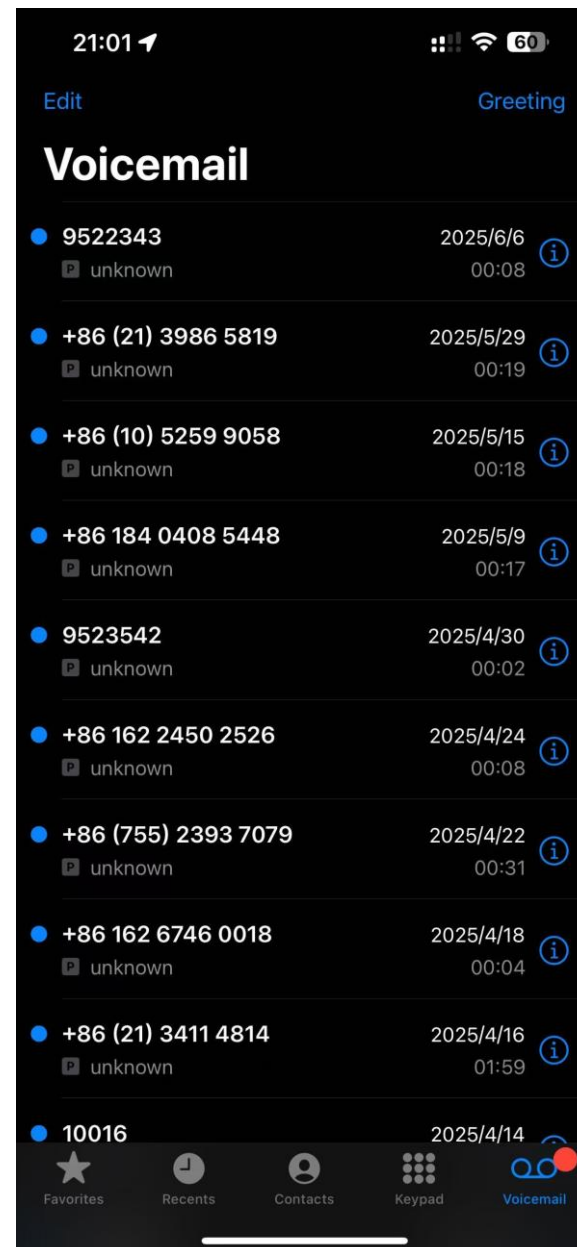
Spatial · AEL REL

Widget Function · Widget Interaction · Action Understanding · Action Prediction  
Absolute Element Location · Relative Element Location

Qwen2.5-VL-7B-Instruct · InternVL2.5-8B | parsed generation correctness | parsing failure = 0

# Dimensions

WF · Widget function

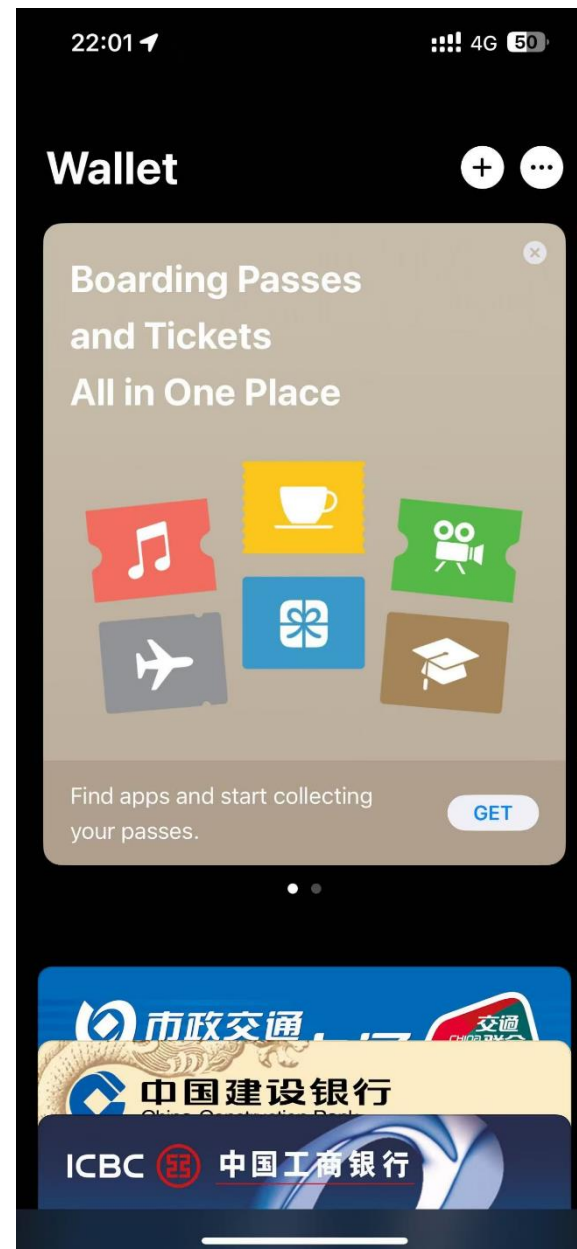


Based on the visual layout of the Voicemail tab in this mobile UI screenshot, what is the function of the small blue "i" icon found beside each call entry?

- A. Edit the contact information associated with the voicemail.
- B. Open detailed information about the voicemail, including playback options.**
- C. Mark the voicemail as read or unread.
- D. Add the phone number to contacts.

# Dimensions

WI · Widget interaction



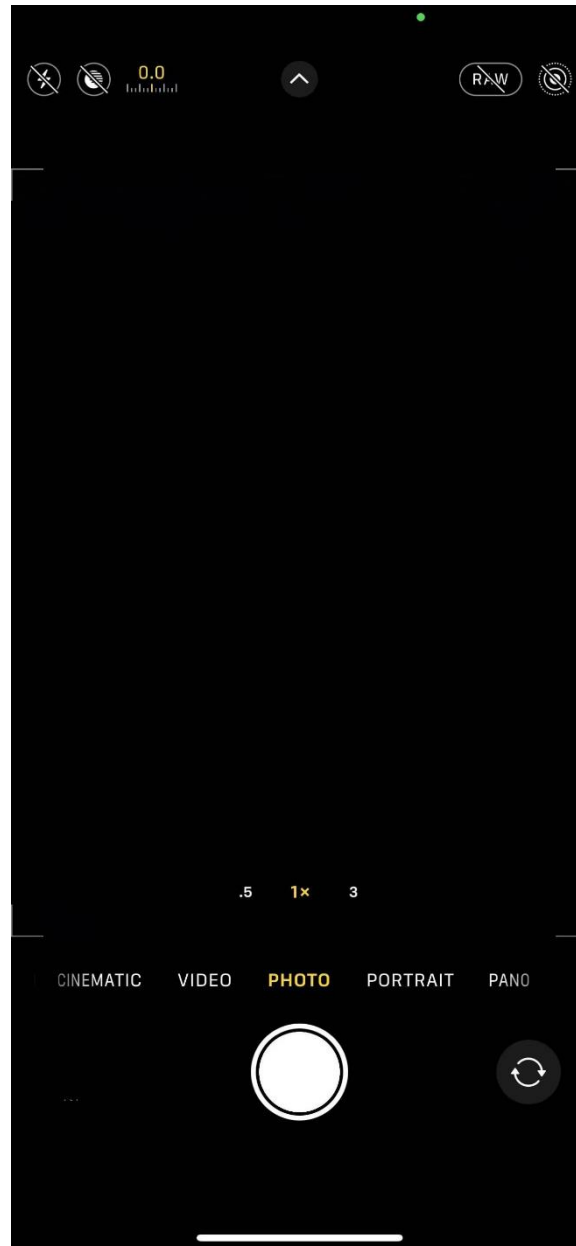
Which action will initiate the process to download a pass-related app in this mobile interface screenshot?

- A. Tap the "GET" button in the bottom-right corner.
- B. Press the plus icon at the top-right corner.
- C. Swipe horizontally on the card collection at the bottom.
- D. Click the "Boarding Passes and Tickets" text.



# Dimensions

AU · Action understanding

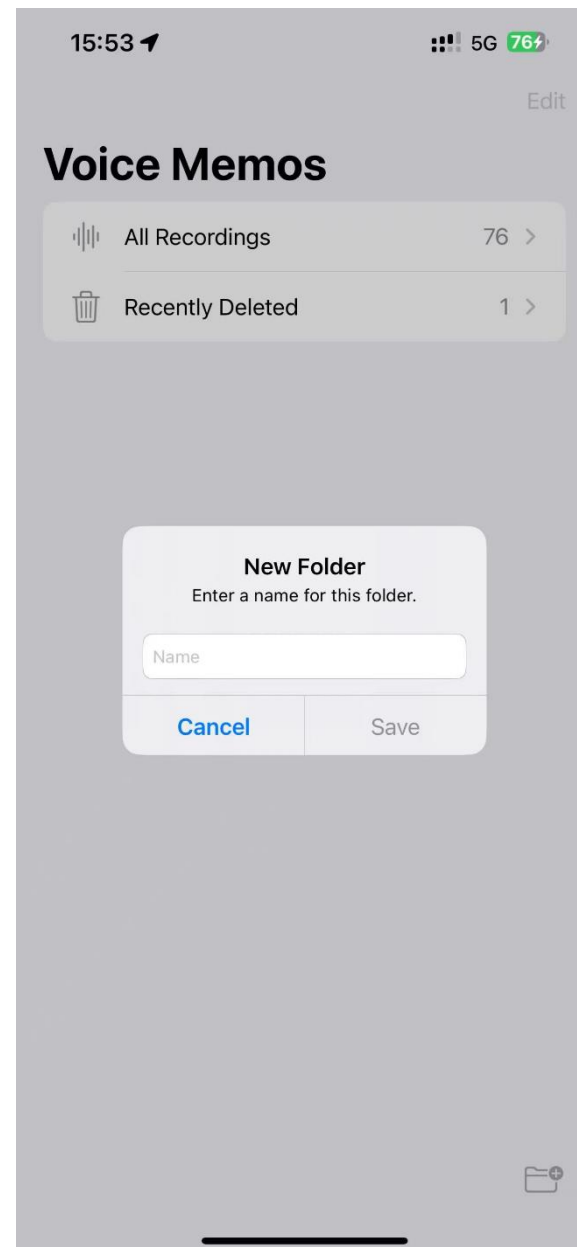


What immediate effect will occur if the user presses the shutter button in this camera interface?

- A. The camera will switch to video mode.
- B. A photo will be taken, and the preview will appear on the screen.**
- C. A popup asking to save the photo will appear.
- D. The screen will navigate to the settings menu.

# Dimensions

AP · Action prediction

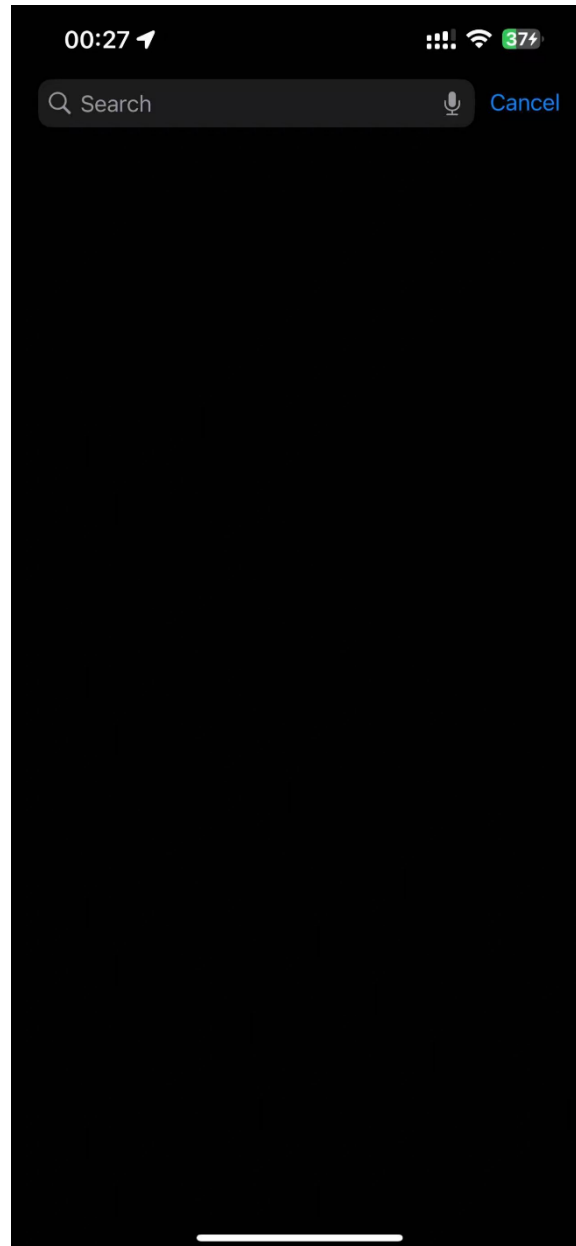


The user wants to create a new folder named "Work Ideas" in the Voice Memos app. Which sequence is correct?

- A. Tap "Cancel" → tap the Name field → type "Work Ideas" → tap "Save".
- B. Tap the Name field → type "Work Ideas" → tap "Save".**
- C. Tap "Edit" → tap "All Recordings" → tap the Name field → type "Work Ideas".
- D. Tap "Save" → tap the Name field → type "Work Ideas".

# Dimensions

AEL · Absolute location



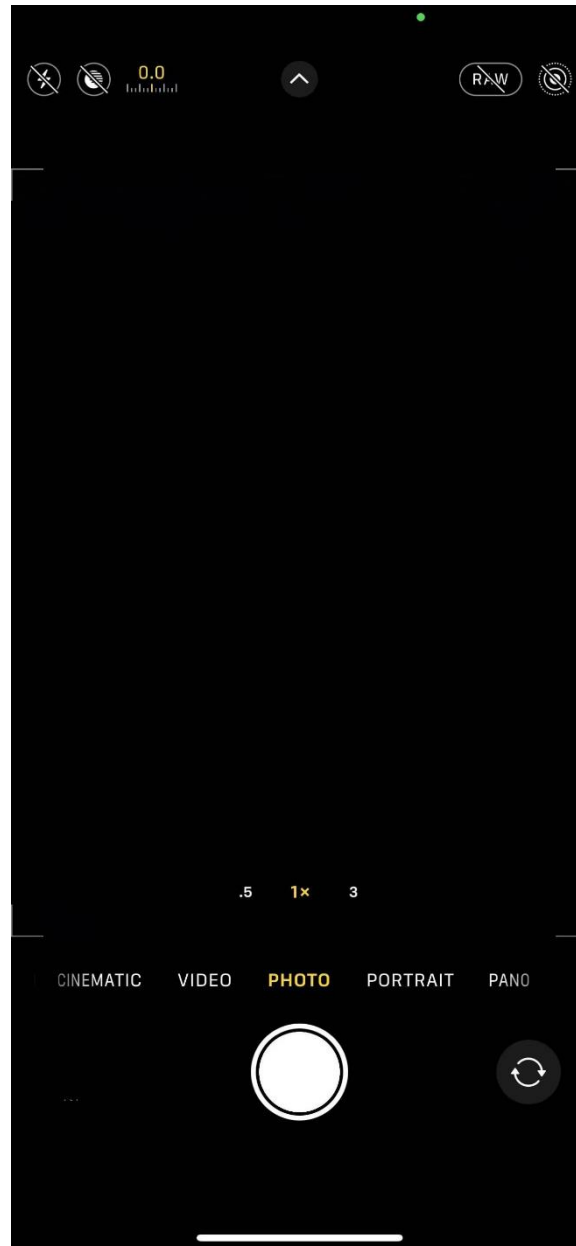
In which area of the screen is the "Cancel" button located?

- A. Top-right corner.
- B. Bottom-right corner.
- C. Top-left corner.
- D. Bottom-left corner.



# Dimensions

REL · Relative location



Where is the Photo mode option located relative to the rest of the mode options on the bottom toolbar?

- A. To the far right of all mode options.
- B. To the far left of all mode options.
- C. In the center, between "Video" and "Portrait".**
- D. Just to the left of "Pano" mode.

# Separating Language Difficulty from Concordance Advantage

Primary GLMM + secondary GEE robustness

$$\text{logit}(p_{ij}) = \beta_0 + \mathbf{x}_{ij}^{(Q)} \beta_Q + \mathbf{x}_{ij}^{(G)} \beta_G + \mathbf{x}_{ij}^{(D)} \beta_D + \beta_M M_{ij} + b_i$$

$p_{ij}$ : correct-answer probability    $M_{ij}$ : matched indicator    $b_i$ : item-specific baseline difficulty

## PRIMARY

### Logistic GLMM

- Query-language difficulty
- GUI-language difficulty
- Dimension difficulty
- Item-specific baseline difficulty
- Matched condition의 추가 이점

correct ~ query\_language + gui\_language + dimension + matched + (1 | parallel\_id)

## ROBUSTNESS

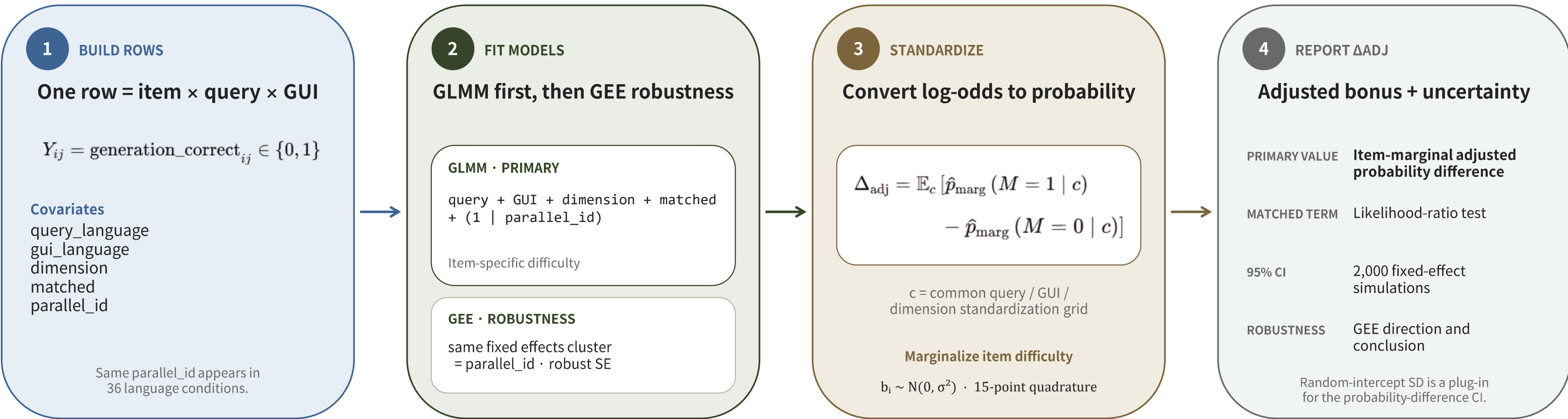
### Logistic GEE

- Same-item observations = one cluster
- Exchangeable within-item correlation
- Robust standard errors
- Population-averaged association
- Different framework, same research question

GLMM은 item difficulty를 직접 모델링하고, GEE는 item-level correlation을 robust하게 처리함

# Separating Language Difficulty from Concordance Advantage

How the Adjusted Concordance Bonus Is Estimated

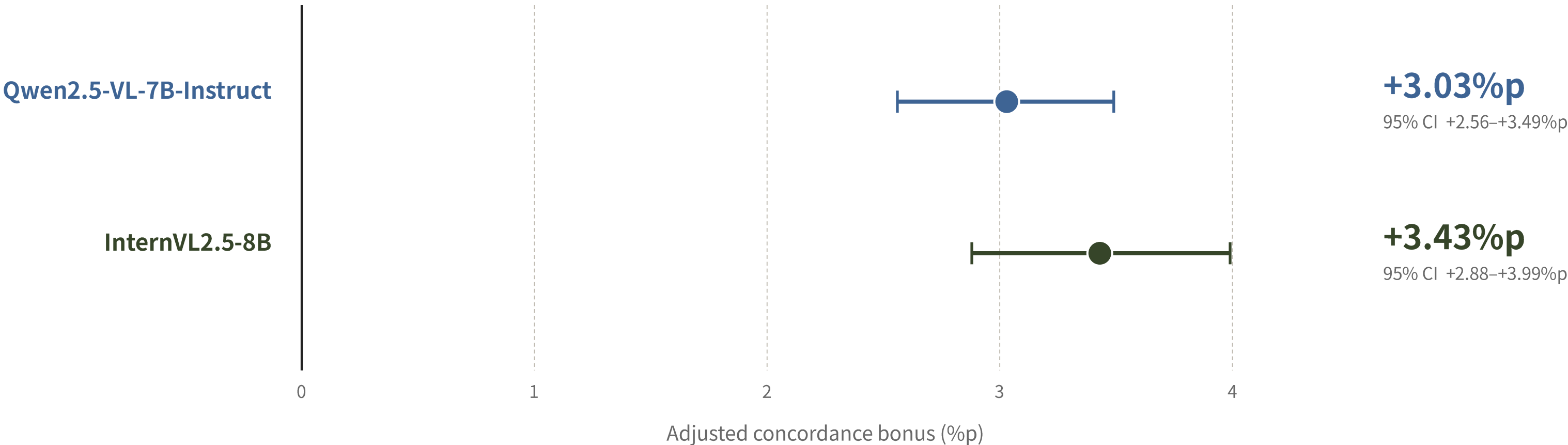


Raw gap은 관찰된 차이이고, adjusted bonus는 언어 · dimension · item 난이도를 고려한 뒤 남는 concordance-associated difference다.

NOT A RAW DIAGONAL-OFF-DIAGONAL ACCURACY GAP

# Language Concordance Improves Accuracy After Adjustment

RQ1 · item-marginal GLMM adjusted probability difference

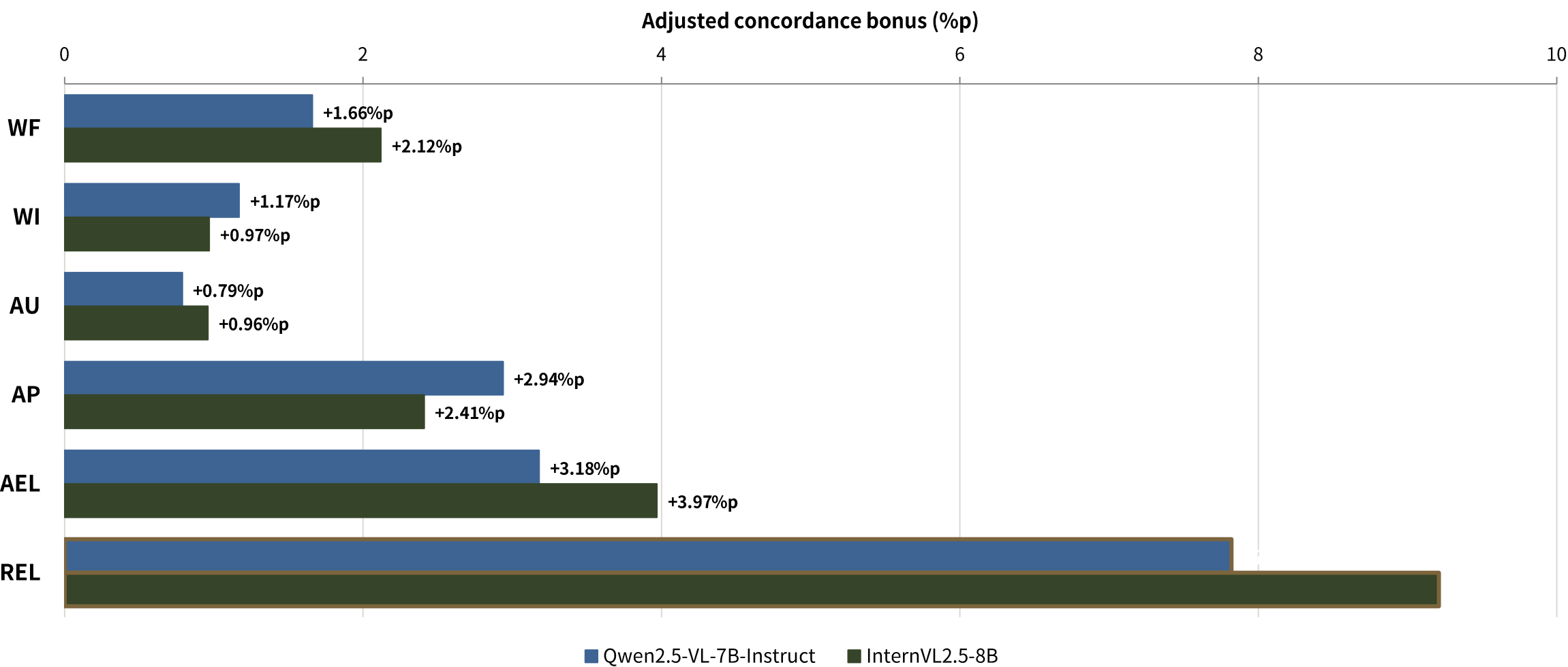


Positive adjusted concordance advantage was observed in both models.

GEE robustness showed the same positive direction · Qwen +3.41%p | InternVL +3.83%p

# Concordance Advantage Is Largest in Relative Element Location

RQ2 · adjusted concordance bonus by GUI capability



MATCHED × DIMENSION

Significant in both models

REL showed the largest bonus

**Qwen +7.82%p**

**InternVL +9.21%p**

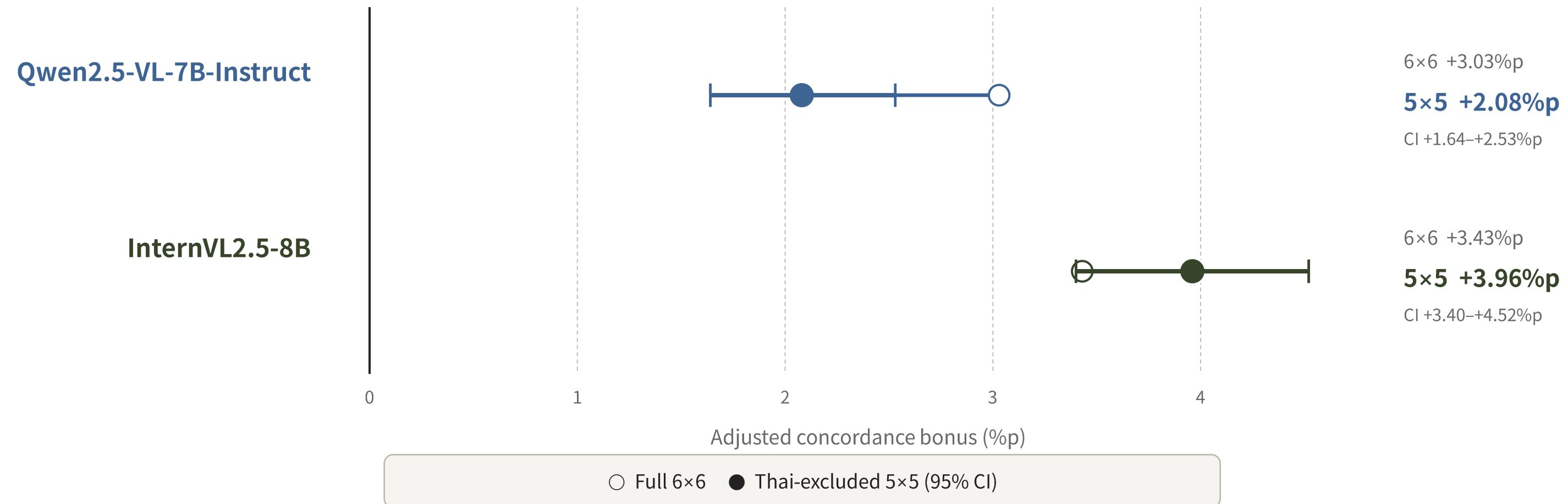
AU · no clear advantage  
WI · small, model-dependent  
AP / AEL · intermediate

Omnibus matched × dimension LRT · follow-up trends used Holm correction



# The Global Concordance Effect Is Not Driven Only by Thai

Robustness · refitting the global model on a 5×5 subset

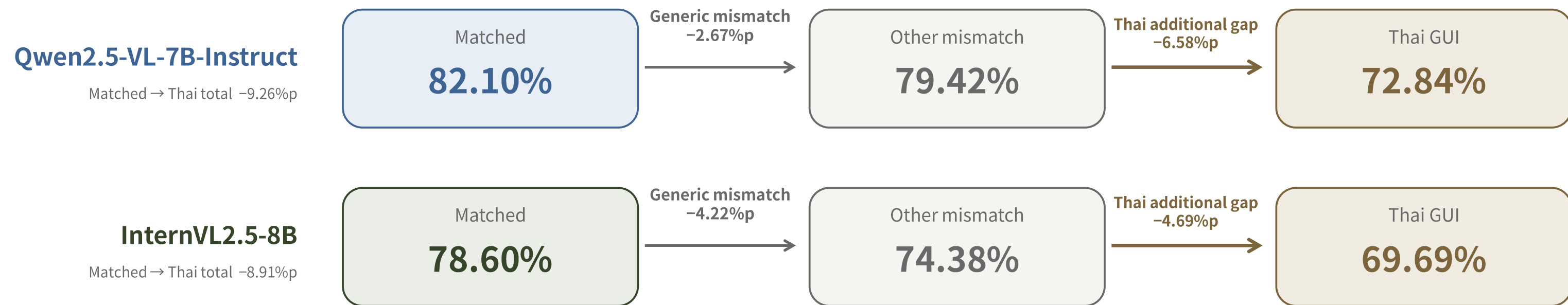


Removing Thai from both query and GUI conditions did not eliminate the adjusted concordance advantage.

5×5에서는 global RQ1만 재검정했으며, REL ranking은 별도의 interaction model로 직접 검정하지 않았다.

# Thai GUI Produces an Additional Raw Drop, Especially in REL

Secondary diagnostic observation · paired raw contrast



MATCHED → THAI GUI TOTAL DROP IN REL

**Qwen -21.42%p**

**InternVL -25.14%p**

REL의 total drop에는 generic mismatch와 Thai additional raw gap이 함께 포함됨

RAW PAIRED CONTRAST — NOT AN ADJUSTED THAI-SPECIFIC GLMM EFFECT

# Cross-locale GUI Understanding Requires Capability-level Diagnosis

Interim conclusion and next steps

## Key findings

- 1 Global concordance advantage**  
Qwen +3.03%p · InternVL +3.43%p
- 2 Capability heterogeneity**  
Relative Element Location (REL)에서 가장 큰 adjusted difference
- 3 Thai-excluded robustness**  
5×5에서도 global effect 유지
- 4 Thai diagnostic pattern**  
Thai GUI additional raw drop · REL에서 특히 큼

## Next steps

- 1 REL / AEL 실패 사례 annotation**
- 2 Text-dependent vs. text-light item 분리**
- 3 Query translation intervention**
- 4 GUI text translation / replacement**
- 5 Thai-specific adjusted GLMM if needed**
- 6 Option-order robustness**

The current result diagnoses where mismatch matters; the next step is to identify why and test targeted mitigation.